G3SP : Generator of Synthetic Scans of Spectroscopic Plates for Object Detection

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Abstract. Spectroscopic analysis has a long tradition in astronomy, where stellar spectra were historically recorded on photographic plates. These plates constitute a valuable scientific and historical archive, enabling the study of long-term variability in astronomical objects. However, they are prone to degradation and information loss, which has motivated large-scale digitization efforts.

Despite ongoing initiatives, many plates remain undigitized, and the digitization process itself is time-consuming. Moreover, several downstream tasks required for scientific analysis—such as the detection and localization of spectral observations—are still performed manually. This step is particularly critical, as it is both labor-intensive and error-prone, creating a bottleneck in the data processing pipeline.

A major limitation for automating these tasks using modern computer vision methods is the scarcity of labeled data. In this work, we present G3SP (Generator of Synthetic Scans of Spectroscopic Plates), a tool designed to generate realistic, labeled scans of spectroscopic plates. The generated images incorporate domain-specific characteristics and simulated defects, such as dust, moisture artifacts, and illumination variations, while providing precise annotations for each observation.

To evaluate the usefulness of the proposed approach, we train an object detection model under three settings: using only real data, synthetic data, and their combination. Our results show that synthetic data generated with G3SP capture the underlying visual structure of spectroscopic plates and provide a strong basis for model generalization. In particular, models trained on synthetic data achieve competitive performance and further improve when combined with real data, reaching up to 0.995 mAP_{50} and 0.95 mAP_{50-95}.

Keywords: Generated data, Object detection, Spectroscopic plates, Synthetic data

1 Spectroscopic Data

Spectroscopy constitutes a fundamental tool for the physical study of astronomical objects. Through the analysis of electromagnetic radiation received, absorbed, or reflected by an object, it is possible to infer several of its physical properties. For example, spectroscopy can reveal the chemical composition, temperature, and radial velocity of astronomical sources. This technique enables the study of celestial bodies beyond their position or apparent brightness in the sky [5].

In the 20th century, the acquisition of spectroscopic data using photographic techniques was widespread. This method consisted of recording spectra through photography, with the distinctive feature that the images were stored on glass plates rather than on photographic paper. The plates captured during this time have high historical and scientific value, as they contain information about astronomical objects in the past, some of which are highly valuable, particularly those corresponding to southern hemisphere objects, which were historically less observed than those of the northern hemisphere during that period. Today, many of these plates are preserved in their original institutions [7]. Figure 1 shows a spectroscopic glass plate containing two observations on its surface.

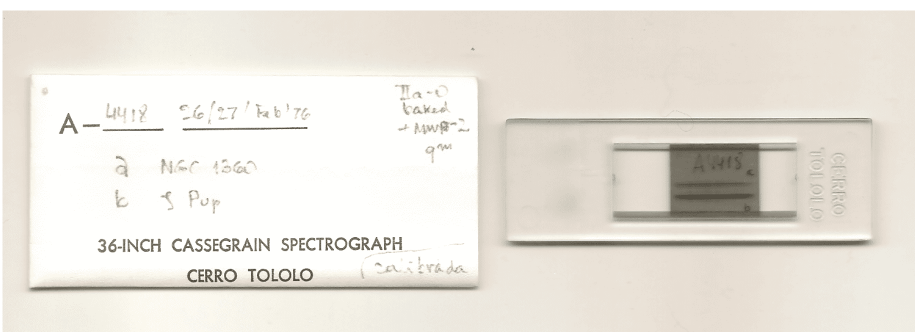
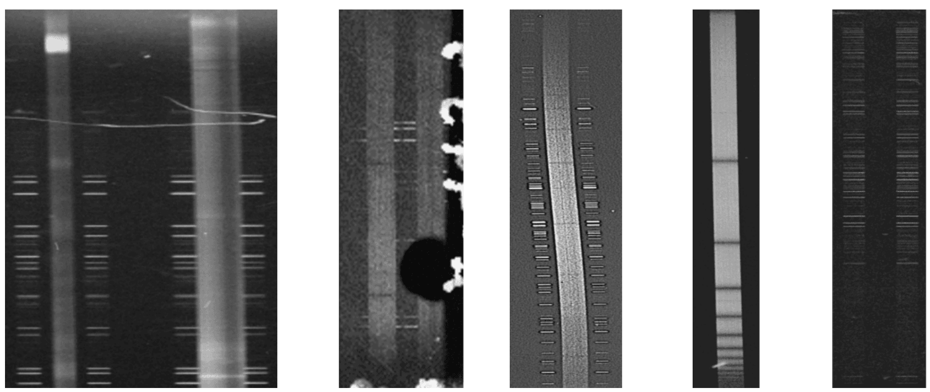


Fig. 1: Left, the wrapping includes additional information about each observation. Right, spectroscopic plate `A4418` of the astronomical source `Zeta Puppis`, captured at the Observatorio Interamericano de Cerro Tololo (Chile) as part of the `ReTrOH` [11] project. The plate contains two spectroscopic observations.

However, storing information in physical plates can present challenges under suboptimal storage conditions. Several incidents affecting this type of heritage have occurred. In 2016, due to a burst pipe, the Harvard College Observatory suffered a flood that affected more than 61,000 plates in its archive, including part of the Harvard plate collection currently digitized by the DASCH project [21]. A similar situation occurred at the Facultad de Ciencias Astronómicas y Geofísicas of the Universidad Nacional de La Plata, where mold was discovered on the furniture used to store the plates [13]. The risk of mold, dust, scratches, or other types of damage is significant and, if not properly addressed, can lead to the permanent degradation of the information stored on each plate.

In addition to physical deterioration, plates may present defects associated with the data acquisition process, such as curvature in the spectral trace, imaging artifacts, or the absence of certain calibration elements. Figure 2 shows examples of plates exhibiting observations with different types of defects.



(a) (b) (c) (d) (e)

Fig. 2: Samples of possible defects on the plates. Defects associated with prolonged storage: (a) Scratches. (b) Dirt and/or moisture stains. Defects associated with the capture process: (c) Curved engraving. (d) Missing comparison lamps. (e) Comparison lamps without their respective spectrum.

To preserve this heritage, institutions such as the International Astronomical Union (IAU) have issued resolutions[5] highlighting the importance of preserving spectroscopic plates for future generations of astronomers. The IAU recommends digitization as a conservation strategy. This process enables the creation of multiple copies and redundant archives, helping to preserve the most relevant information in the long term.

[5] IAU Resolution B3 (August 2000): Safeguarding the Information in Photographic Observations

To address these preservation challenges, many institutions have launched digitization projects for their photographic plate collections. Table 1 lists some of the most relevant collections containing large numbers of spectroscopic plates, together with their digitization status and calibration progress.

Table 1: Some relevant projects or institutions with collections of spectroscopic plates. (a) It is known that HCO has a collection of spectroscopic plates, but the exact number is unknown.

Collection	#plates.	Digitized	Calibrated	Date range	References
`HCO`	?[(a)]	?%	0%	1880–1992	[21]
`APPLAUSE`	19.240	100%	0%	1893–1999	[3]
`ReTrHO`	>15.000	4%	0%	1920–1980	[13]
`WFPDB`	?	100%	?%	1858–2005	[19,20]
`OHP Archive`	63.500	100%	?%	1944–1997	[15]
`Foster Archive`	>4.800	100%	?%	1928–1991	[2]
`DFBS`	1.874	100%	100%	1965–1980	[10]

Plates need to be digitized and calibrated in order to be useful for scientific work. Digitization consists of scanning the plate and storing its information in a digital format. Although the process presents several challenges, different approaches exist to accomplish this, either using adapted scanners [9,12] or specialized hardware [15,18,17]. On average, scanning a plate takes approximately 15–30 minutes. Calibration involves extracting relevant features from the plate and usually consists of several steps with varying levels of complexity. In previous work, we presented PlateUNLP, a software tool designed to assist in the calibration process. This tool can speed up calibration using machine learning, signal processing, and metadata-based approaches [1]. In this work, we improve one stage of the calibration process, which consists of detecting observations in plate scans.

2 Detection problem

An observation consists of the spectrum of an astronomical object together with two symmetrical comparison lamps located above and below it. A single plate can contain many observations; however, to extract scientific data, each observation must be isolated in a separate file. Figure 3 shows an example of how the regions corresponding to each observation should be defined. Each observation is defined as a bounding box enclosing the spectrum and its corresponding comparison lamps.

In previous work, we presented a YOLO-based detector [16] that automates observation detection in spectroscopic plates. The model was trained exclusively on real scan images applying data augmentation. The model achieved good per-

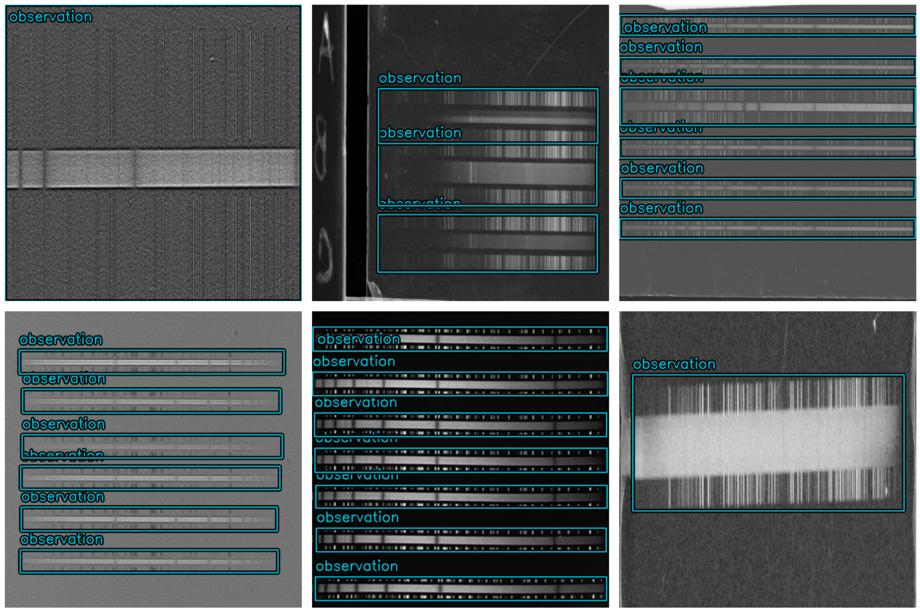


Fig. 3: The images show examples of real spectroscopic plates. The bounding boxes enclose the regions corresponding to each observation, which consist of the spectrum and its corresponding comparison lamps. If the observation is too large, the delimiting box extends to the edge of the scan, as in the leftmost top image.

formance for this detection task but showed limited precision when predicting the exact boundaries of observations.

The predicted bounding boxes frequently misalign with the true observation boundaries, which can negatively impact downstream calibration steps. In particular, incomplete observation detections may lead to inaccuracies in wavelength calibration, thereby compromising the analysis of the extracted spectral information.

2.1 Evaluation metrics

One metric that highlights this behavior is the mAP at different IoU thresholds. IoU (intersection over union, Equation 1) measures the overlap between the predicted bounding box and the ground truth bounding box. Its values range from 0 (no overlap) to 1 (perfect overlap).

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} \tag{1}$$

The mean Average Precision (mAP) considers a detection to be correct if its IoU is greater than or equal to a given threshold. Based on this criterion, a precision-recall curve is constructed, and the area under this curve defines the Average Precision (AP) (Equation 2). When multiple classes are involved, AP is computed for each class and then averaged to obtain the mAP (Equation 3). mAP values range from 0 to 1, with higher values indicating better detection and localization performance [4,8].

$$AP = \int_0^1 p(r)dr \tag{2}$$

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i \tag{3}$$

Implementations of YOLO, such as Ultralytics [6], report validation metrics during training. Among these, mAP at different IoU thresholds is included by default. For example, mAP_{50} measures the model's general ability to detect objects, while mAP_{50-95} evaluates its performance under stricter localization requirements, reflecting its ability to produce more precisely aligned bounding boxes.

2.2 Explanation of precision problem

In previous work, we observed that the model trained on real data achieved good performance in detecting the presence of observations, but showed limited precision in fitting their boundaries. This behavior was evidenced by a significant drop in mAP as the IoU threshold increased. The model performed well in terms of mAP_{50}, but poorly in mAP_{50-95}.

We attribute this behavior to the small number of scans available for training. A model trained on a limited dataset tends to memorize features, which limits its ability to precisely localize observation boundaries in new images. In the original work, data augmentation techniques were applied to mitigate this issue. However, since the initial dataset is very small, the generated variations remain limited. Consequently, the precision problem persists and becomes particularly evident when evaluating mAP_{50-95}.

A straightforward solution to the small dataset problem is to acquire more data. However, as mentioned before, the digitization process is time-consuming. Combined with the lack of adequate hardware and software tools, many institutions choose not to digitize their plates. This further restricts the amount of data that can be obtained in most cases.

In this work, we present a tool to mitigate the issue of limited training data. A procedural generator of synthetic spectroscopic plate scans is proposed, which enables the creation of large datasets with precise annotations. The generated images capture the underlying visual structure of spectroscopic plates, providing a strong basis for generalization for training object detection models. It enables training with small datasets while maintaining high detection performance for observations with precise positions and dimensions.

3 G3SP

G3SP[6] is a software tool designed to generate synthetic spectroscopic plate scan data. The generated images preserve a realistic appearance and allow the inclusion of artificial defects such as dust stains, moisture marks, reflections, among others. Each generated image is accompanied by labels containing the coordinates and shapes of each observation.

The tool enables fine-grained control over the generated data, such as observation geometry, observation layout, spectral characteristics, and the noise patterns. Some of the parameters that can be configured are shown in Table 2.

All parameters are defined as ranges, ensuring that the generated observations exhibit the desired variability. The produced observations preserve the structural characteristics of spectroscopic plates. For example, the comparison lamps located on both sides of each science spectrum are always generated symmetrically. The generated images can also reproduce extreme cases that are present in real plates but are rare, such as plates with many observations, plates with observations that are very close to each other, or plates with a single observation occupying almost the entire plate. This variability allows the model to learn the general visual patterns of this type of plate, providing a strong basis for generalization for subsequent fine-tuning. Figure 4 shows examples of generated plate scans.

The generated labels follow the `rel_xywh` format, according to the YOLO annotation scheme [14]. This means that the coordinates of each observation are expressed as normalized values between 0 and 1.

```
<class_id> <x_center> <y_center> <width> <height>
```

<class_id> represents the detection class. Since only one class (observations) is considered, its value is always 0. *<x_center>* and *<y_center>* correspond to the normalized coordinates of the bounding box center, while *<width>* and *<height>* represent its normalized dimensions. All values except *<class_id>* are expressed in the range $[0, 1]$. Figure 4 shows examples of these annotations.

G3SP can be installed as a package. The library provides several functions for generating synthetic data directly from code. Users may either call these functions manually or use the implemented generator object, which is compatible with TensorFlow pipelines. The generator internally relies on the same functions, making both approaches effectively equivalent.

3.1 Available Data

For this study, we use a dataset composed of 361 spectroscopic plates from the archive of the Facultad de Ciencias Astronómicas y Geofísicas of the Universidad Nacional de La Plata belonging to the recovery project ReTrOH[7]. The dataset

[6] G3SP repository

[7] Project ReTrOH: Recovery of Historical Observational Work

Table 2: Main parameters of G3SP for synthetic spectroscopic plate generation.

Parameter	**Description**
Canvas and Global Properties	
height_range, width_range	Range of canvas dimensions.
gray_value_range	Background gray level.
Observation Geometry	
obs_width, obs_height	Size of each observation relative to the canvas.
angle_range	Inclination of observations.
opening_lamp_range	Relative size of lamp spectra.
distance_between_components_range	Spacing between spectral components.
Observation Distribution	
distance_between_observations_range	Spacing between observations.
cant_observations_max	Maximum number of observations.
noise_horizontal, noise_vertical	Positional perturbations.
target filtering	Random removal of observations (10% probability).
Spectral Modeling	
n_peaks	Number of emission peaks.
n_absorption_lines	Number of absorption lines.
baseline	Base intensity level.
peak_spread	Width of emission peaks.
absorption_lines_spread	Width of absorption lines.
vertical_noise_level	Vertical intensity perturbation.
Noise and Artifacts	
gaussian_std_range	Gaussian noise level.
band_intensity_range	Horizontal band noise.
speck_count_range, speck_size_range	Dust artifacts.
blur_kernel_size_options	Blur kernel size.
prob_edge	Plate edge artifact probability.
violin_line_count	Number of elongated artifacts.
violin_intensity	Intensity of elongated artifacts.
violin_length_range	Length of elongated artifacts.
Output Configuration	
resize_shape	Output resolution.
batch_size	Batch size.
max_predictions	Maximum annotations per image.
output_format	Output label format.

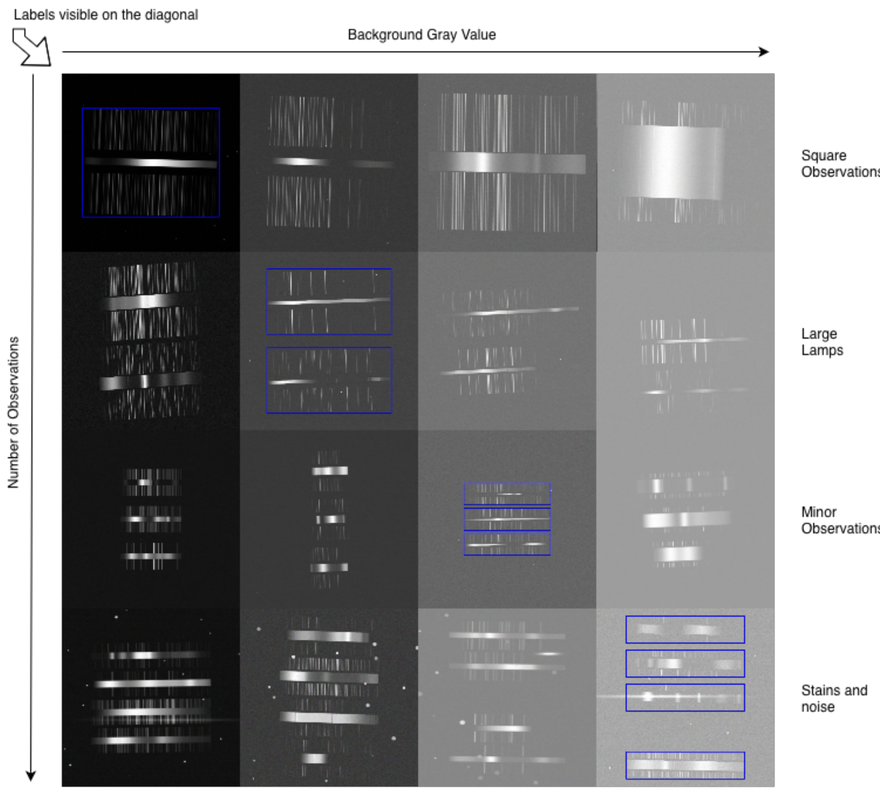


Fig. 4: Generated plate scans. Rows vary in characteristics such as observation count, shape, and noise. Background gray level increases from left to right, and the number of observations increases from top to bottom. Diagonal images (top-left to bottom-right) show the corresponding labels.

is publicly available[8]. Since the dataset is relatively small, an additional 96,000 synthetic images were generated using G3SP to complement it.

Each dataset was divided into training and test sets using an 80/20 split:

- Real images (361): 289 for training and 72 for testing.
- Generated images (96,000): 76,800 for training and 19,200 for testing.

4 Detection model

For this task, we adopted the YOLO architecture [14], specifically the lightweight variant *yolo_v26_n* implemented in recent versions of the Ultralytics library [6]. This variant is designed for real-time object detection, prioritizing computational efficiency while maintaining competitive performance.

[8] Observation Detection in Spectroscopic Plates Dataset

Training was performed using a batch size of 32 and an input image resolution of 640×640 pixels. Optimization parameters were automatically selected, with an initial learning rate of 0.01, a final learning rate factor of 0.01, momentum of 0.937, and weight decay of 0.0005. A warmup phase of 3 epochs was applied.

The number of training epochs varied depending on the dataset: models trained on synthetic data were trained for 20 epochs, while those trained on real data were trained for 500 epochs. This difference is justified by the larger volume of synthetic samples compared to real ones.

Data augmentation included HSV color variations, random scaling (±50%), horizontal flipping (50% probability), translation (10%), and mosaic augmentation enabled during early training.

The training process leveraged automatic mixed precision (AMP) and was executed on an *NVIDIA GeForce RTX 3090* GPU with 24 GB of VRAM, ensuring efficient computation and memory usage.

The trained model allows up to 300 detections per image. Unlike traditional object detectors, the end-to-end architecture of *yolo_v26_n* eliminates the need for non-maximum suppression (NMS), as redundant detections are resolved internally by the model.

4.1 Experiments

Five training evaluations were conducted to evaluate the performance of the proposed approach. Each experiment involved training a model under different training and testing datasets.

An initial model was trained using only real data to establish a baseline performance. This model was then evaluated on both real and generated datasets. If the performance is significantly worse on generated data, it indicates that the problem is more difficult than what is reflected in the training data, or that the limited amount of real data does not provide sufficient information for the model to generalize to new samples. If the performance is close to zero, it suggests that the generated data differs substantially from the real data.

With the baseline established, a second model was trained using only generated data and evaluated on both real and generated datasets. This experiment allows us to assess the model's generalization capability when trained on synthetic data and applied to real images, as well as its ability to capture patterns present in the generated data.

Finally, a third model was trained using a combination of real and generated data. Specifically, the model was first trained on generated data and subsequently fine-tuned using real data. This approach aims to leverage the strengths of both datasets, providing a more diverse training set that captures the variability present in real plates while benefiting from the large volume of synthetic samples. The performance of this model was evaluated only on real data. As an additional experiment, we also evaluated the last model on the entire real dataset (real training set and real testing set).

Regarding data augmentation, it was applied to all models, including those trained on generated data. For synthetic data, augmentation is less necessary, as the generator already produces a wide variety of samples and the number of training epochs is relatively small, which reduces repetition. However, applying augmentation still helps introduce additional variability in the few samples that are repeated. In contrast, augmentation is essential when training on real data, as the dataset is very limited and the model tends to overfit easily without it. Additionally, mosaic augmentation is particularly useful, as the random composition of images resembles certain configurations observed in real spectroscopic plates. While not specific to this dataset, it reflects realistic scenarios.

To measure test performance, the mAP at different IoU thresholds was used (mAP_{50} and mAP_{50-95}). This metric allows us to evaluate both the model's ability to detect the presence of observations and its precision in fitting their boundaries. Table 3 shows the metrics obtained in each evaluation. Figure 5 shows example predictions obtained with the model trained on generated data, on real data and trained on generated data and fine-tuned using a limited set of real images.

Table 3: Results with real, generated and mixed training data.

training data	test data	epochs	mAP_{50}	mAP_{50-95}
real	*generated*	500	0.662	0.419
real	*real*	500	0.992	0.895
generated	*generated*	20	0.995	0.99
generated	*real*	20	0.92	0.755
generated+real	*real*	20+297	**0.995**	**0.95**
generated+real	*real (train+test)*	20+297	0.995	0.966

4.2 Analysis

The first two evaluations in Table 3 show that the model trained on real data achieves excellent performance in detecting and localizing observations, with a general detection score (mAP_{50}) of 0.992 and a more stringent metric (mAP_{50-95}) of 0.895. However, when the same model is evaluated on generated data, its performance drops significantly, reaching 0.662 and 0.419, respectively.

This result indicates that the model is overfitting to the limited real training data. Since both the training and testing sets originate from the same real data distribution, they share many characteristics, which facilitates high performance despite overfitting. In contrast, the notable degradation when evaluated on generated data, which follows a different distribution, highlights the poor generalization capability of models trained exclusively on real data. Although data augmentation introduces some variability, it is clearly insufficient to ensure robust generalization.

The evaluation of the model trained exclusively on generated data shows a different behavior. When evaluated on real data, its performance decreases compared to the model trained on real data, achieving an mAP_{50} of 0.92 and an

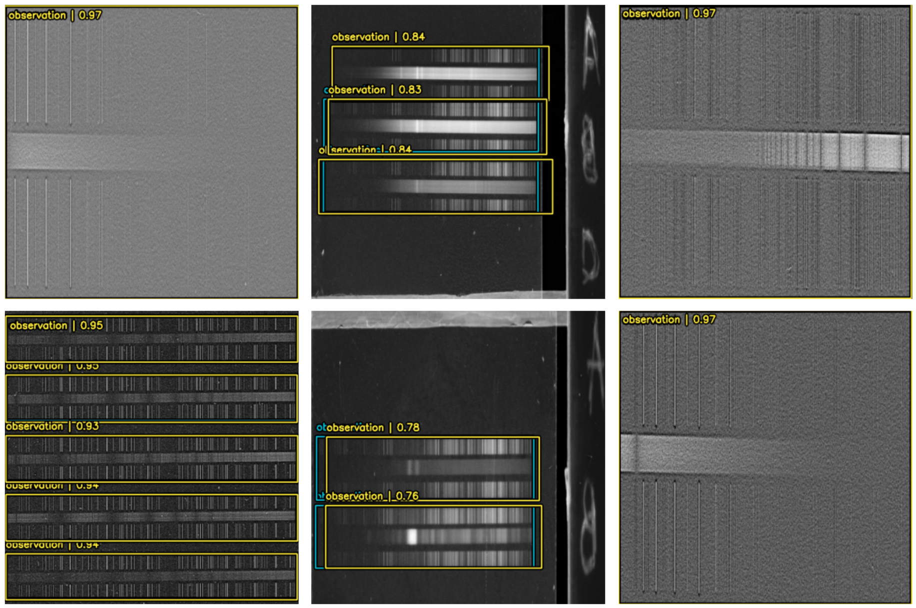


Fig. 5: Predictions obtained with the model trained on generated data and fine-tuned using a limited set of real images. Blue bounding boxes correspond to the ground truth, while red bounding boxes represent the model's predictions. The model achieves good performance in both detecting the presence of observations and accurately fitting their boundaries.

mAP_{50-95} of 0.755. However, its performance does not degrade when evaluated on data following the same synthetic distribution. On the contrary, the model achieves excellent results on generated data, with scores of 0.995 and 0.99 for mAP_{50} and mAP_{50-95}, respectively. This is expected, as the model was trained on synthetic data. However, the fact that it maintains strong performance on both datasets, despite not being trained on real data, indicates a good generalization capability. This is a desirable characteristic that the baseline model does not exhibit.

The final evaluation considers the model trained on generated data and subsequently fine-tuned using real training data. This approach combines the generalization capability acquired from synthetic data with the ability to capture the complex patterns present in real images, patterns to which the first model overfits. This strategy achieves the best overall performance, with an mAP_{50} of 0.995 and an mAP_{50-95} of 0.95, clearly outperforming the other two models. These results demonstrate the effectiveness of using synthetic data for pretraining, followed by fine-tuning on real data, to achieve superior performance not only in detecting observations but also in accurately localizing them.

5 Conclusions

Given the good performance of models based on data generated with G3SP, we can conclude that the pretraining of the model with generated data for observation detection in spectroscopic plates is a viable approach for achieving excellent performance and good generalization capability. This option is particularly relevant in scenarios where the availability of real training data is limited, as is the case with spectroscopic plates. The generated data provides a strong generalization basis, enabling the model to learn the underlying visual patterns of this type of plate. Moreover, fine-tuning with real data allows the model to capture complex patterns present in real images, further improving its performance. In this sense, the performance achieved in previous works using only augmented real data was outperformed by this approach.

The synthetic spectroscopic plate scan generator presented in this work, G3SP, demonstrates—through the performance of models trained on its data—its ability to capture part of the complex patterns associated with this type of spectroscopic images. By generating realistic and diverse samples, the approach can be generalized and adapted to a wide range of spectroscopic scan scenarios. Its inherent randomness enables models to better handle plates containing multiple observations with varying spectral positions and shapes.

In future work, we plan to enhance the generation capabilities of G3SP by producing more realistic images with increasingly complex patterns and a wider variety of defects. We also intend to extend the training and annotation process beyond observation detection to include the precise identification of individual components within each observation. This would represent a step forward in the automation of the full calibration process of spectroscopic plates, which remains a bottleneck in the processing pipeline of this type of historical, valuable, and vulnerable scientific heritage data.

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